

UPGMA TREE

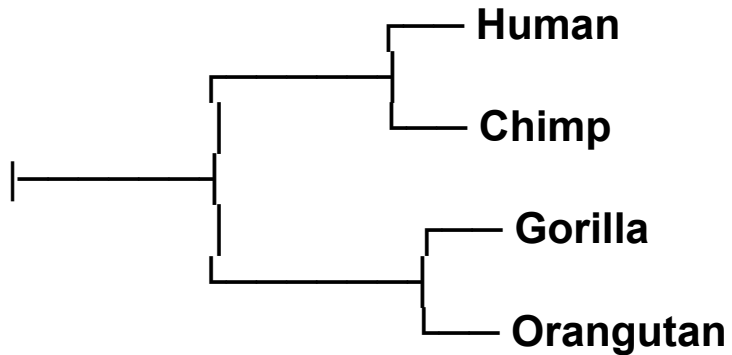
FROM SEQUENCE

TO DISTANCE MATRIX

Species	Sequence		Human	Chimp	Gorilla	Orangutan
Human	ATGCTAGC	Human	0			
Chimp	ATGCTTGC	Chimp	0.125	0		
Gorilla	ATGATAGC	Gorilla	0.125	0.250	0	
Orangutan	TTGATAGC	Orangutan	0.250	0.375	0.125	0

UPGMA TREE

	HC	Gorilla	Orangutan			
HC	0			HC	HC	GO
Gorilla	0.1875	0		HC	0	0.25
Orangutan	0.3125	0.125	0	GO	0.25	0



CLASS WORK

	S1	S2	S3	S4	S5	S6
S1	0.000000	1.989113	1.993307	1.951477	2.073501	2.055488
S2	1.989113	0.000000	0.825802	0.550602	0.465545	0.826270
S3	1.993307	0.825802	0.000000	0.693857	0.666949	0.489254
S4	1.951477	0.550602	0.693857	0.000000	0.353383	0.689581
S5	2.073501	0.465545	0.666949	0.353383	0.000000	0.665220
S6	2.055488	0.826270	0.489254	0.689581	0.665220	0.000000

<i>Homo sapiens</i>	S5
<i>Ustilago maydis_521</i>	S6
<i>Mimivirus</i>	S1
<i>Anopheles</i>	S2
<i>Schizosaccharomyces pombe 972h</i>	S3
<i>Strongylocentrotus_purpuratus</i>	S4

Robust phylogenetic analysis for everyone.

► Free, simple to use web service dedicated to reconstructing and analysing phylogenetic relationships between molecular sequences.

► **Let's GO ! with One Click Workflow**



► One Click

Fully automatic workflow
Default tools + **default** parameters.



► Advanced

Semi automatic workflow
Default tools + **custom** parameters



► A la Carte

Custom workflow
Custom tools + **Custom** parameters.

DATA Preparation

>SEQ1

AGCTCGTAATAGCGCGAATATAAGCGTACA

>SEQ2

AGCTCGTAATAGTGCGAATATAAGCGTACA

>SEQ3

AGCTCGTAATAGCGCGAAGATAAGCGTACA

>SEQ4

AGCTCATAATAGCGCGAATATAAGCGTACA

>SEQ5

AGCTCGTAATAGCGCGAATATAAGCGTACA

>SEQ6

AGCTCGTAATAGCGCGAATATAAGNNTACA

One Click Workflows

Select your workflow

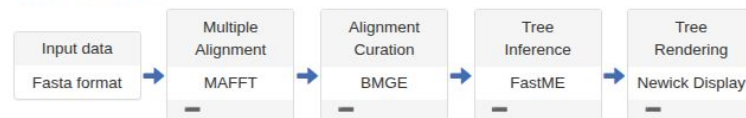
FastME/OneClick

PhyML/OneClick

PhyML+SMS/OneClick

FastTree/OneClick

Workflow skeleton



Choose input data

Input Data (Fasta format with more than 3 sequences)

Input file

Choose file To Upload

Choose File

Pasted text

```
>SEQ2
AGCTCGTAATAGTGC GAATATAAGCGTACA
>SEQ3
AGCTCGTAATAGCGCGAAGATAAGCGTACA
>SEQ4
AGCTCATAATAGCGCGAATATAAGCGTACA
>SEQ5
AGCTCGTAATAGCGCGAATATAAGCGTACA
>SEQ6
AGCTCGTAATAGCGCGAATATAAGNTACA
```

Blast run*

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Galaxyfile*

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Submit

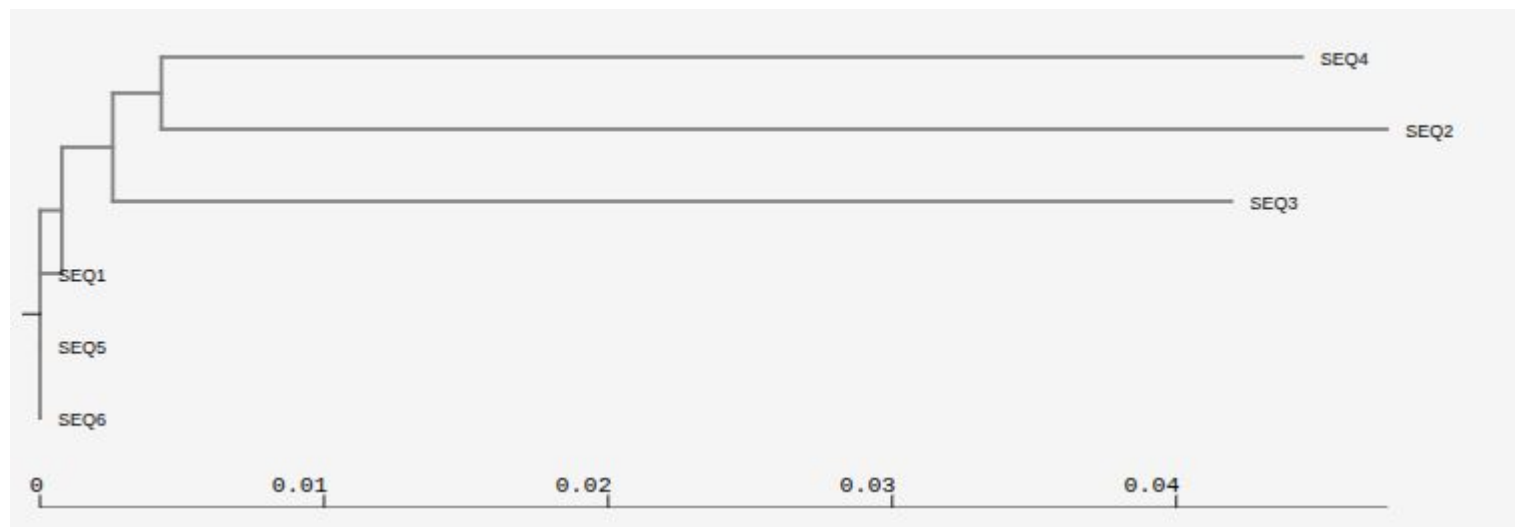
Example



DISTANCE MATRIX

6						
S1	0.000000	0.045536	0.040549	0.043089	0.000000	0.000000
S2	0.045536	0.000000	0.084881	0.083326	0.045536	0.049057
S3	0.040549	0.084881	0.000000	0.080683	0.040549	0.043224
S4	0.043089	0.083326	0.080683	0.000000	0.043089	0.046179
S5	0.000000	0.045536	0.040549	0.043089	0.000000	0.000000
S6	0.000000	0.049057	0.043224	0.046179	0.000000	0.000000

TREE VIEWER



- Download or copy-paste the FASTA sequences into a .fasta file.
- Construct phylogenetic tree using <https://ngphylogeny.fr/>
[Data here](#)

- Download or copy-paste the FASTA sequences into a .fasta file.
- Construct phylogenetic tree using <https://ngphylogeny.fr/>

NR_024570.1

NR_112116.1

NR_113956.1

NR_074828.1

NR_025911.1

NR_026142.1

<https://www.phylogeny.fr/>

Phylogeny.fr

A comprehensive, simple-to-use platform for phylogenetic analysis

[One Click Analysis](#)

[Learn More](#)



Ready-to-use Pipelines

Run pre-configured pipelines for fast and accurate phylogenetic analysis.



Customizable Workflows

Advanced users can choose from a-la-carte tools to tailor their workflow.



Result Visualization

Generate high-quality phylogenetic trees with downloadable formats.

DATA Preparation

>SEQ1

AGCTCGTAATAGCGCGAATATAAGCGTACA

>SEQ2

AGCTCGTAATAGTGCGAATATAAGCGTACA

>SEQ3

AGCTCGTAATAGCGCGAAGATAAGCGTACA

>SEQ4

AGCTCATAATAGCGCGAATATAAGCGTACA

>SEQ5

AGCTCGTAATAGCGCGAATATAAGCGTACA

>SEQ6

AGCTCGTAATAGCGCGAATATAAGNNTACA

1

Alignment
Mafft

2

Curation
ClipKit

3

Phylogeny
FastTree

One Click Phylogeny

Upload a FASTA file:



Drag and drop your file here

or click to browse (.faa, .fa, .fasta, .txt, .aln)

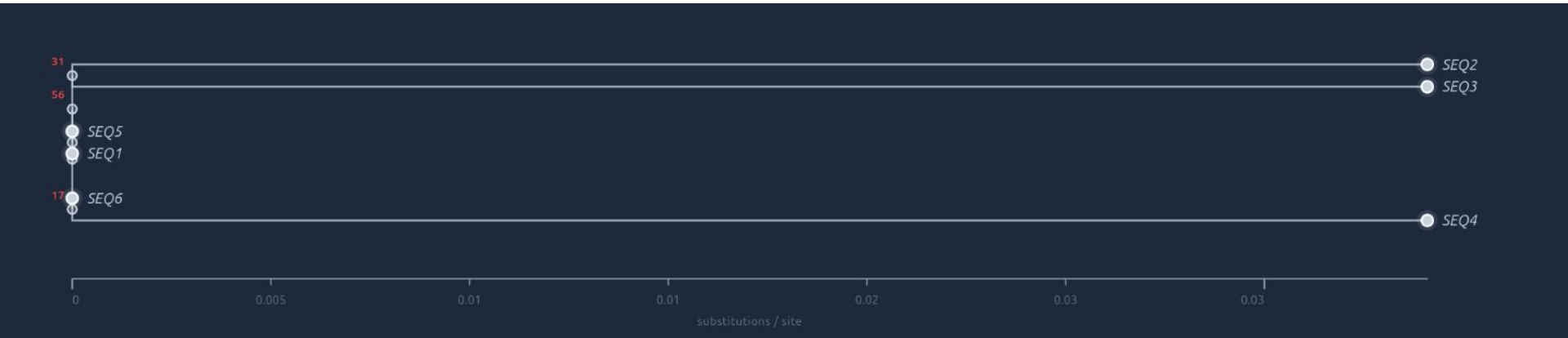
Or paste sequences here:

```
>SEQ1
AGCTCGTAATAGCGGAATATAAGCGTACA
>SEQ2
AGCTCGTAATAGTGCGAATATAAGCGTACA
>SEQ3
AGCTCGTAATAGCGGAAGATAAGCGTACA
>SEQ4
AGCTCATAATAGCGGAATATAAGCGTACA
>SEQ5
AGCTCGTAATAGCGGAATATAAGCGTACA
```

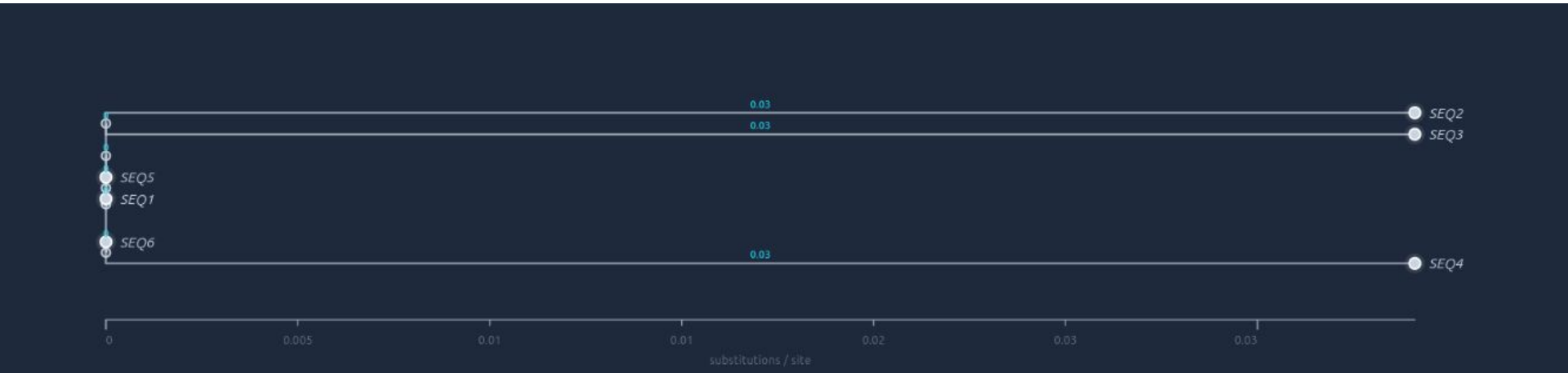
Example

[Need to convert from another format? \(Nexus, Phylip...\)](#)

BOOTSTRAP



BRANCH LENGTH



1. Go to <https://www.phylogeny.fr>
2. Click "One Click" mode (or "Advanced" if you want to customize)

3. Use FASTA format for all:

NC_012920.1

NC_005089.1

NC_001643.1

NC_001645.1

NC_002082.1

NC_002083.1

4. Run and wait for:

- Multiple Sequence Alignment
- Tree construction
- Bootstrap analysis

>Homo_sapiens
GTGCCAGCAGCCGCGGTAATACGGCACATGATTTGGTGACACGGTACCTGAGGAGTGACGGGCGGTGTGTACAAAGGGCAGGGACGTTTACCCACCAA
CTAGAGGAGCCTGTTTACCAAAAACATCACGGTCAAGGTGAAATTTGGAAGGTGACAGAGTGACAAGCCCTAGGGATAACAGCGCAATCAGTTAAAAAGTT
AAAGGCGGGGAAGCAGGCGGCTTTATTTGTAGCTGGTTTTAAAGTTGGGGTAGATGTAAGGGA

>Mus_musculus
GTGCCAGCAGCCGCGGTAATACGGCACACGATTTCTGTGAAAGGTGCCCTGAGGAGTGACGGGCGGTGTGTACAAAGGGCAGGGACGTTTACCCACCAA
CTAGAGGAGCCAGTTTACCAAAAACATCACGGTCAAGGTGAAATTTGGAAGGTGACAGAGTGACAAGCCCTAGGGATAACAGCGCAATCAGTTAAAAAGTT
AAAGGCGGGGAAGCAGGCGGCTTTATTTGTAGCTGGTTTTAAAGTTGGGGTAGATGTAAGGGA

>Pan_troglodytes
GTGCCAGCAGCCGCGGTAATACGGCACATGATTTCTGTGAAAGGTACCTTGAGGAGTGACGGGCGGTGTGTACAAAGGGCAGGGACGTTTACCCACCAAC
TAGAGGAGCCTGTTTACCAAAAACATCACGGTCAAGGTGAAATTTGGAAGGTGACAGAGTGACAAGCCCTAGGGATAACAGCGCAATCAGTTAAAAAGTTA
AAGGCGGGGAAGCAGGCGGCTTTATTTGTAGCTGGTTTTAAAGTTGGGGTAGATGTAAGGGA

>Gorilla_gorilla
GTGCCAGCAGCCGCGGTAATACGGCACATGATTTCTGTGAAAGGTACCTTGAGGAGTGACGGGCGGTGTGTACAAAGGGCAGGGACGTTTACCCACCAAC
TAGAGGAGCCTGTTTACCAAAAACATCACGGTCAAGGTGAAATTTGGAAGGTGACAGAGTGACAAGCCCTAGGGATAACAGCGCAATCAGTTAAAAAGTTA
AAGGCGGGGAAGCAGGCGGCTTTATTTGTAGCTGGTTTTAAAGTTGGGGTAGATGTAAGGGA

>Hylobates_lar
GTGCCAGCAGCCGCGGTAATACGGCACATGATTTCTGTGAAAGGTACCTTGAGGAGTGACGGGCGGTGTGTACAAAGGGCAGGGACGTTTACCCACCAAC
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AAGGCGGGGAAGCAGGCGGCTTTATTTGTAGCTGGTTTTAAAGTTGGGGTAGATGTAAGGGA

>Pongo_abelii
GTGCCAGCAGCCGCGGTAATACGGCACATGATTTCTGTGAAAGGTACCTTGAGGAGTGACGGGCGGTGTGTACAAAGGGCAGGGACGTTTACCCACCAAC
TAGAGGAGCCTGTTTACCAAAAACATCACGGTCAAGGTGAAATTTGGAAGGTGACAGAGTGACAAGCCCTAGGGATAACAGCGCAATCAGTTAAAAAGTTA
AAGGCGGGGAAGCAGGCGGCTTTATTTGTAGCTGGTTTTAAAGTTGGGGTAGATGTAAGGGA

>Escherichia_coli
AGAGTTTGGATCCTGGCTCAGGATGAACGCTGGCGGCGTGCCTAATACATGCAAGTCGAACGGTGAAGGTTTTCGGATCGTAAAGCTCTGTTGTTAAGGAAGAACAAGTACGCTGACCTTACAGATGG
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GAGACACGGTCCAGACTCCTACGGGAGGCAGCAG

>Bacillus_subtilis
AGAGTTTGATCCTGGCTCAGGATGAACGCTGGCGGCGTGCCTAATACATGCAAGTCGAACGGTGAAGGTTTTCGGATCGTAAAGCTCTGTTGTTAGGGAAGAACAAGTACGCTGACCTTACAGATG
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TGAGACACGGTCCAGACTCCTACGGGAGGCAGCAG

>Staphylococcus_aureus
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GACCCGCGGCGCATTAGCTAGTTGGTGAGGTAATGGCTACCAAGGCGACGATCCGTAACCTGGTCTGAGAGGATGATCAGTCACACTGGAACGGTACCTGAGAGGGTGATCGGCCACACTGGAAC
TGAGACACGGTCCAGACTCCTACGGGAGGCAGCAG

>Pseudomonas_aeruginosa
AGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTCGAGCGGATGAAGGTTTTCGGATCGTAAAGCTCTGTTGTTAGGGAAGAACAAGTACCGTAGGCTTACAGAT
GGACCCGCGGCGCATTAGCTAGTTGGTGAGGTAATGGCTACCAAGGCGACGATCCGTAACCTGGTCTGAGAGGATGATCAGTCACACTGGAACGGTACCTGAGAGGGTGATCGGCCACACTGGAAC
CTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAG

>Lactobacillus_acidophilus
AGAGTTTGATCCTGGCTCAGGATGAACGCTGGCGGCAGGCCTAACACATGCAAGTCGAACGGTAACAGGAAGCAGCCTGCCGATAAGGGAAGCCAGGGGAAACTGGGTCTAATACCGGATAATGA
CCAGATAGCGAAGAACCGGATCCGCCCACGGTGAATACGTTCCCGGGCCTTGATACACACCGCCCGTCACACCATGGGAGTGGGTTGCACCAGAAGTAGCTAGTCTAACCTTCGGGAGGACGGTT
ACCACGGTGTGATGGGAGTGGGTTGCACCAGAAG

>Clostridium_difficile
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GACCCGCGGCGCATTAGCTAGTTGGTGAGGTAATGGCTACCAAGGCGACGATCCGTAACCTGGTCTGAGAGGATGATCAGTCACACTGGAACGGTACCTGAGAGGGTGATCGGCCACACTGGAAC
TGAGACACGGTCCAGACTCCTACGGGAGGCAGCAG

>Salmonella_enterica
AGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCGTGCCTAATACATGCAAGTCGAACGGTGAAGGTTTTCGGATCGTAAAGCTCTGTTGTTAAGGAAGAACAAGTACGCTGACCTTACAGATGGA
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AGACACGGTCCAGACTCCTACGGGAGGCAGCAG

>Mycobacterium_tuberculosis
AGAGTTTGATCCTGGCTCAGGATGAACGCTGGCGGCAGGCCTAACACATGCAAGTCGAACGGTAACAGGAAGCAGCCTGCCGATAAGGGAAGCCAGGGGAAACTGGGTCTAATACCGGATAATGA
CCAGATAGCGAAGAACCGGATCCGCCCACGGTGAATACGTTCCCGGGCCTTGATACACACCGCCCGTCACACCATGGGAGTGGGTTGCACCAGAAGTAGCTAGTCTAACCTTCGGGAGGACGGTT
ACCACGGTGTGATGGGAGTGGGTTGCACCAGAAG

>Vibrio_cholerae
AGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTCGAACGGTGAAGGTTTTCGGATCGTAAAGCTCTGTTGTTAGGGAAGAACAAGTACCGTAGGCTTACAGATG
GACCCGCGGCGCATTAGCTAGTTGGTGAGGTAATGGCTACCAAGGCGACGATCCGTAACCTGGTCTGAGAGGATGATCAGTCACACTGGAACGGTACCTGAGAGGGTGATCGGCCACACTGGAAC
TGAGACACGGTCCAGACTCCTACGGGAGGCAGCAG

>Streptococcus_pneumoniae
AGAGTTTGATCCTGGCTCAGGATGAACGCTGGCGGCAGGCCTAACACATGCAAGTCGAACGGTGAAGGTTTTCGGATCGTAAAGCTCTGTTGTTAAGGAAGAACAAGTACCGTAGGCTTACAGATGG
ACCCGCGGCGCATTAGCTAGTTGGTGAGGTAATGGCTACCAAGGCGACGATCCGTAACCTGGTCTGAGAGGATGATCAGTCACACTGGAACGGTACCTGAGAGGGTGATCGGCCACACTGGAAC
GAGACACGGTCCAGACTCCTACGGGAGGCAGCAG